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Rumors Stop with the Wise: Unveiling Inbound SAV Deployment through Spoofed ICMP Messages

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Abstract

In the era of increasing network-based threats, particularly IP spoofing, Source Address Validation (SAV) is paramount for network security. The effective deployment of Inbound Source Address Validation (ISAV) is crucial yet often inadequate, posing significant risks to Internet infrastructure. This study presents *ICMP_Sonar*, a measurement system that deploys "rumors"—carefully crafted spoofed ICMP packets—to probe the network's defenses, revealing the "wise" networks with their robust ISAV implementations. *ICMP_Sonar* introduces two novel approaches that exploit the characteristics of ICMP unreachable messages and ICMP fragment needed messages, and exhibits the advantages of high coverage, fine granularity, low error rates, and the ability to measure in both IPv4 and IPv6. We also evaluate the applicability and security risks of ICMP error messages. Through large-scale measurements, *ICMP_Sonar* successfully covers 86M IPv4 hosts (0.8M IPv6 hosts), 3.5M IPv4 /24 subnets (24K IPv6 /40 subnets), and 59K IPv4 ASes (8.3K IPv6 ASes), surpassing the state-of-the-art dual-stack method's coverage by 16.2 (51.6), 2.9 (2.34), and 1.7 (1.7) times, respectively. The broad coverage across multiple granularities enables us to capture a more comprehensive and fine-grained view of ISAV deployment. Measurements show that while the percentage of ASes with no ISAV deployment is lower than previously identified, the percentage of ASes with partial ISAV deployment is much higher, indicating significant gaps in overall security. The analysis also reveals that ISAV deployment practices vary across different networks and between IPv4 and IPv6.

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CCS Concepts

• **Security and privacy → Network security**.

Keywords

Network Measurement; Spoofing; Source Address Validation

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1 Introduction

IP spoofing [15] is a prevalent threat in large-scale Distributed Denial of Service (DDoS) attacks [9][17][32][33]. With IP spoofing, attackers manipulate the source address in packet headers to mask their identity and location, facilitating anonymity and hindering traceability. For years, Source Address Validation (SAV) has emerged as a critical mechanism designed to prohibit attackers from using spoofed source addresses that fall outside the range of legitimately advertised prefixes.

RFC 2827 (BCP 38) [13] and RFC 3704 (BCP 84) [6] formalize the practice and urge Internet Service Providers (ISPs) to implement SAV. Additionally, SAV is recommended in the Mutually Agreed Norms for Routing Security (MANRS) [4] initiative, which was launched by the Internet Society (ISOC) [3] to improve the security and resilience of the global routing system.

SAV can be categorized into Outbound SAV (OSAV) and Inbound SAV (ISAV) [22]. A network deploying OSAV discards outbound packets with source addresses not belonging to the network, while a network deploying ISAV drops inbound packets with source addresses that belong to the network. OSAV plays an important role in preventing a network from being the source of an attack, while ISAV is crucial for protecting the network against incoming threats.

Despite the significance of SAV, its implementation has not been satisfactory [22], which may be hindered by lack of awareness of the spoof threat [18]. Besides, some network administrators are

concerned that deploying SAV may inadvertently impact legitimate network traffic, particularly in complex scenarios such as multi-homing [6][36]. Nevertheless, deploying SAV remains the best strategy for mitigating IP spoofing attacks. The absence of SAV compromises the integrity and trustworthiness of network communications. In addition to DDoS attacks [38], it may also cause security issues, such as spoofing attackers being able to access services that are only available for local clients [37].

Measuring the SAV deployment of individual networks and alerting insecure networks to deploy SAV as soon as possible is an important way to prevent spoofing attacks. However, the deployment of SAV, particularly ISAV, is difficult to measure due to the lack of Vantage Points (VP) and generalized measurement methods. Although existing SAV measurement efforts have provided valuable insights into the deployment and effectiveness of SAV across the Internet, they face challenges such as reliance on voluntary participation, insufficient coverage on IPv4 and IPv6 networks, few measurable targets in large networks, and potential errors. Addressing these challenges is crucial for developing a more comprehensive and accurate understanding of global SAV implementation.

This study introduces two novel approaches implemented in *ICMP_Sonar*, a tool using meticulously crafted spoofed ICMP messages to detect the presence and efficacy of ISAV. Our *ICMP_Sonar* relies on the feature that ICMP unreachable messages can prevent TCP retransmissions and the feature that ICMP fragment needed messages can force the target to reply in fragments, offering high coverage across networks (59K/8.3K IPv4/IPv6 ASes and 3.5M/24K IPv4/IPv6 subnets), fine granularity in finding a large number of measurable hosts (86M IPv4 hosts and 0.8M IPv6 hosts), low error rates, and the ability to measure in both IPv4 and IPv6.

Through large-scale measurement, the advantages of our methods have led to new discoveries: The extensive coverage provided by our dual-stack measurements enables us to identify significant differences in ISAV deployment across networks of varying hierarchies, types, and geographical distributions. Additionally, ISAV deployment in IPv6 is notably less comprehensive than in IPv4. Moreover, the wealth of measurable targets within networks has highlighted both the prevalence and limitations of subnet-level ISAV deployment strategies: although 60% of IPv4 ASes and 46% of IPv6 ASes have deployed ISAV, only 11% of IPv4 ASes and 13% of IPv6 ASes fully protect all of their subnets. Importantly, many critical resources, such as top domains, remain unprotected. The variability in ISAV deployment across different subnets indicates that the overall security of ASes is not impenetrable. Existing research often considers the security situation of ISAV across the Internet based on the AS-level deployment rate, which could lead to an overestimation of security situation of ISAV on the Internet.

In this paper, we have made the following contributions:

(1) We propose two novel dual-stack methods, namely the ICMP unreachable method and the ICMP fragmentation method, for measuring the deployment of ISAV in the target network without vantage points. We also conduct local experiments on the applicability of our methods.
(2) We validate the effectiveness and feasibility of these two methods through large-scale measurement experiments. We also analyze the conditions under which the error may occur and demonstrate the low error probability of our two measurement methods. The measurement results demonstrate the advantages of our methods in terms of dual-stack coverage and granularity.
(3) We analyze the measurement results and find that a significant portion of ISAV is deployed at subnet boundaries, and there are substantial differences in ISAV deployment across IPv4 and IPv6, as well as among different networks.
(4) We conduct a survey to investigate the actual deployment of ISAV on the Internet, and the results confirm our measurement findings.

By mapping the landscape of ISAV deployment, we aspire to empower network administrators, policymakers, and cybersecurity professionals with the knowledge to fortify their defenses against the ever-evolving threats in cyberspace. This paper delves into the methodologies, findings, and implications of our research, providing a comprehensive examination of ISAV's role in securing the Internet infrastructure.

2 Related Work

Table 1 summarizes the existing measurement work on SAV deployment, which can be divided into two approaches based on whether VPs need to be deployed within target networks.

One approach is to deploy VPs within the target network to send or receive spoofed packets and detect the deployment of SAV. For years, the Spoofer project [5][7][18][22] has contributed to recruiting volunteers or renting workers from crowd-sourcing platforms [20] to run the Spoofer client, which can be used to detect whether it receives the spoofed packets that originate from the client's network, and thus accurately infer SAV deployment. However, it is challenging to deploy VPs in any network, only about 2.2K IPv4 ASes (0.7K IPv6 ASes) and 12K IPv4 /24 subnets (1.9K IPv6 /40 subnets) can be measured through the project. Although Spoofer provides a lower bound on the number of networks that fail to deploy SAV, it is still difficult to infer the SAV security of the entire Internet.

To overcome the difficulty in deploying VPs, another approach, which uses measurement techniques to remotely detect SAV deployment without the need to deploy VPs in the target network, has become popular in recent years. For OSAV measurements, researchers have developed methods such as examining loops found in traceroute records and AS relationships [21], or using misconfigured DNS forwarders [19][25] within the target network that forward traffic originating from outside the network. Considering that research [22] has shown the lack of ISAV is significantly more widespread than OSAV (with 59.2-67.0% of ASes not filtering inbound IPv4 traffic compared to 11.9-25.2% of ASes not filtering outbound IPv4 traffic), our paper primarily focuses on measuring ISAV deployment.

Research on ISAV deployment has increased significantly in recent years. Nosyk et al.[28] and Deccio et al.[12] utilized open DNS resolvers within the target network for reflection-based measurements. Specifically, the researchers send a spoofed DNS query to the DNS recursive resolver within the target network. If the network has not deployed ISAV, the DNS recursive resolver will receive this

Method	SAV Type	IPv4/v6	Relies on VP	Relies on DNS	Error Rate	Coverage
Spoofer[7][18][22]	both	both	yes	no	very low	low
Traceroute Loop[21]	OSAV	both	no	no	exist	low
DNS Forwarder[17][19][25]	OSAV	both	no	yes	very low	low
DNS Reflection[11][12][28]	ISAV	both	no	yes	very low	medium
IP-ID[11][34]	ISAV	IPv4	no	no	exist	high in IPv4
TCP-based Fragment[11]	ISAV	IPv4	no	no	very low	high in IPv4
ICMPv6 Rate Limiting[29]	ISAV	IPv6	no	no	exist	high in IPv6
ICMP Unreachable	ISAV	both	no	no	very low	high
Ping-based Fragment	ISAV	both	no	no	very low	high

Table 1: Previous research on the measurement of SAV deployment

query and send a DNS query to the authoritative domain server controlled by the researchers. Their method can cover approximately 35k IPv4 ASes (5k IPv6 ASes) and 1.2M IPv4 /24 subnets (11k IPv6 /40 subnets). But this coverage is still not wide enough for the measurement of Internet. Besides, it is not easy to reproduce because of the need to control an authoritative domain server.

Instead of relying on open DNS resolvers, researchers have noticed that the standard protocols employed in the Internet can be also used to infer ISAV deployment. However, despite the contributions of existing approaches, limitations remain in terms of inference accuracy and dual-stack applicability. More specifically, T. Dai et al. [11] combined the DNS reflection method with newly proposed methods based on IP-ID (IP Identifier) and TCP-based fragmentation in "SMap", conducting a measurement of ISAV deployment across IPv4 networks. The IP-ID method takes advantage of the globally incremental IP-ID counter of a host in the target network to detect whether it receives the spoofed packets. However, the use of global IP-ID counters is not widespread today, and noise in target networks may also lead to inaccurate detections. The TCP-based fragmentation method uses a server that sets the Don't Fragment (DF) bit in the IPv4 header in the target network to detect whether it receives spoofed ICMP fragment needed messages, by checking the path MTU and DF bit. These methods, however, are not applicable to IPv6 due to the absence of IP-ID and DF fields in the header of IPv6, whose ISAV deployment strategy and status may differ from those of IPv4 networks. In terms of IPv6 measurement, Pan et al. [29] utilized ICMPv6 rate limiting to detect whether the target network receives the spoofed packets by examining the ping reply rate. This technique also struggles with accuracy in noisy environments and is inapplicable to IPv4, where ICMP rate limiting is not defined.

Besides, granularity is also an important evaluation metric for both methods and analysis in SAV deployment studies. Although the above methods significantly expand measurement coverage at AS-level, they either lack fine-grained measurement methods capable of measuring a sufficient number of measurable target hosts, or fall short in fine-grained analysis and discussion of SAV deployment, which might lead to misunderstandings about the actual deployment of SAV across networks. The significance of fine-grained analysis has been recognized by researchers [34], who pointed out that the SAV techniques may vary among different subnets within an AS. However, the focus of this research is primarily on identifying SAV implementation techniques instead of conducting large-scale SAV deployment measurements.

In this paper, we propose two complementary methods that leverage ICMP to overcome these limitations. Specifically, the methods are the ICMP unreachable method and the ICMP fragmentation method. On the one hand, ICMP has similar definitions under both IPv4 and IPv6, allowing our two methods to support dual-stack measurements. On the other, ICMP is a widely used protocol for network diagnostics, providing us with high coverage and fine granularity.

3 Methodology

In this section, we present the two methods designed to detect ISAV. The applicability and specific application of the two methods will be discussed later in Sections 4 and 5. Our objective is to send packets with spoofed source addresses to hosts within the target network and verify whether the target hosts receive these packets, which indicates the absence of ISAV in the target network.

Our two methods have their own advantages and scope of application. The ICMP unreachable method is supported by a wider range of target operating systems, but requires the targets to accept TCP connections. The ICMP fragmentation method does not require a TCP connection, but only works on a limited range of target operating systems. By employing both methods in a complementary manner, we can cover a wider range of targets, enhancing the overall effectiveness of our measurement.

3.1 ICMP Unreachable Method

We first use the characteristic that ICMP unreachable messages interrupt TCP connections to determine if a spoofed packet will be dropped by the target network.

The ICMP unreachable method of *ICMP_Sonar* is illustrated in Figure 1. Target host B is not protected by ISAV, and target host C is under the protection of ISAV. *ICMP_Sonar* (deployed on Host A) performs identical probing tests on target host B and target host C. The method sends a regular TCP syn packet and waits for the target's syn/ack response. Upon receiving the response, the method promptly sends ICMP unreachable messages with spoofed source address belonging to the target network with the received response as the payload. Target host B will not retransmit because the corresponding connection is aborted by the ICMP unreachable message, and target host C is not affected because the spoofed

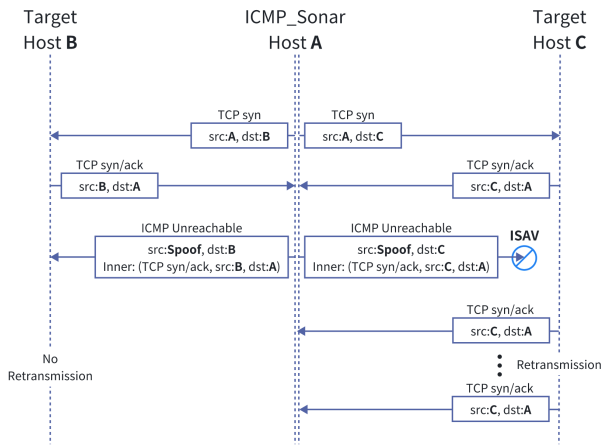


Figure 1: An overview of ICMP unreachable method.

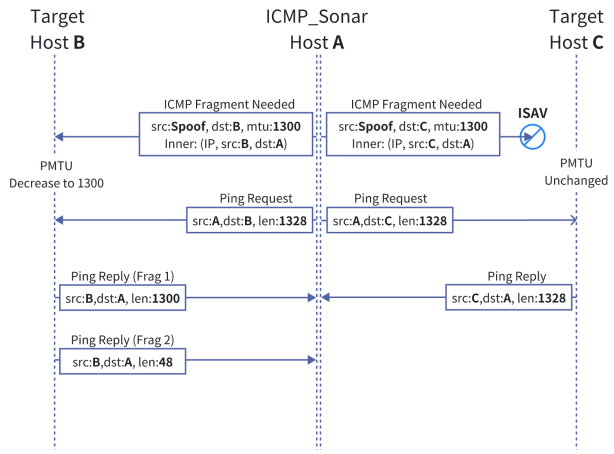


Figure 2: An overview of ICMP fragmentation method.

packet is discarded by ISAV. Therefore, we can determine the ISAV deployment of target host B and target host C by whether they retransmit or not.

ICMP unreachable messages are usually generated by a router or a host to inform the sender that the packet cannot be delivered to its destination address for reasons including network unreachable, host unreachable, no route, administratively prohibited, and so on. The format of ICMP unreachable messages is essentially the same in ICMPv4 [31] and ICMPv6 [10] (see Table 2). The recommended actions for hosts upon receiving such ICMP unreachable messages are also similar, which makes the method applicable to both IPv4 and IPv6 networks. Specifically, the transport layer (e.g. TCP) should respond to ICMP unreachable messages by directing them to the corresponding connection that triggered the error. In RFC 1122 [8], a typical practice for the host is to abort the corresponding connection upon receiving such messages.

ICMP unreachable messages always include the header of the packet that triggered the error as an inner payload. Thus, the target host can match the inner 5-tuple with existing connections, and respond accordingly based on the error type and specific implementation. It is worth noting that the forwarding device that issued the ICMP unreachable message is not a concern for the target host [23]. In other words, the target host typically does not pay attention to the source address of the received ICMP unreachable message, making our measurements feasible. This is expected because, due to the dynamic nature of the Internet, the host finds it challenging to identify whether the outer source IP (source IP of ICMP message) belongs to a router along the forwarding path.

3.2 ICMP Fragmentation Method

In the ICMP fragmentation method, we use the feature that ICMP fragment needed messages can reduce the target PMTU to check whether the target network drops a spoofed packet.

The ICMP fragment needed message has a similar structure to the ICMP unreachable message, as shown in Table 2 (type=3, code=4 in ICMPv4, and type=2, code=0 in ICMPv6). Whether in IPv4 or IPv6, hosts or routers in a network maintain a Maximum Transmission Unit (MTU) value, indicating the maximum size of data service units they can accept. The ICMP fragment needed message is initially defined in RFC 1191 [27] and RFC 1981 [26] for Path MTU Discovery (PMTUD), which is used to find the minimum MTU along an end-to-end path. If a forwarding device along the path has an MTU smaller than the length of the packet, the device will discard the packet and respond with an ICMP fragment needed message, indicating its acceptable MTU value in the "Unused" field of the ICMP message.

Similar to ICMP unreachable messages, ICMP fragment needed messages will also include the header of the received packet that triggered the MTU limitation, allowing the host to identify which packet it sent triggered the MTU restriction. The outer source IP is also ignored by the host. Subsequently, after receiving ICMP fragment needed messages, the host will take action depending on its operating system's specific implementation, such as updating the PMTU value of the specified IP in the routing table cache to the received MTU value. The cache in the routing table is typically set to expire within 10 minutes to avoid a prolonged impact on the host. Not all hosts perform as described above, so we select the measurement targets that perform as desired in Section 5.1.2.

0	1	2 3
Type	Code	Checksum
Unused / Next-Hop MTU		
Internet Header + Part of the Error-raising Package		

Table 2: The format of ICMP unreachable message and ICMP fragmentation message. They are different in type and code. In the ICMP fragment needed message, "Unused" is set to the value of Next-Hop MTU.

To find out whether the PMTU set by a remote host to our host is less than N, which indicates that the target has received the spoofed packet we sent, all we need to do is send a ping request to the target host with a payload length of N bytes. The method does not require any reply to spoofed packets, and is only concerned with whether

the ping replies are fragmented as a result of our measurement. If the PMTU of the target host to our host is set to less than N, its ping reply will be fragmented. Our fragmentation method differs from the one proposed by "SMap" [11] in that: 1. We can measure target hosts that do not accept TCP connections, which broadens the scope of measurement. 2. We do not need to set DF (Don't Fragment in IPv4), thereby gaining the capability to measure in IPv6.

The ICMP fragmentation method of *ICMP_Sonar* is illustrated in Figure 2. The method sends an ICMP fragment needed message with spoofed source address in the target network and MTU field set to 1300. A ping request with a length of 1328 bytes (20 bytes IP header + 8 bytes ICMP header + 1300 bytes payload) is sent a few seconds later. If the target network has not deployed ISAV (Target host B), the ping reply may fragment because the host's PMTU was decreased earlier. Otherwise, we would receive an unfragmented ping reply because the earlier spoofed ICMP fragment needed message was discarded by ISAV (Target host C).

4 Applicability of ICMP Unreachable and Fragment Needed Messages

In this section, we discuss the applicability and security risks of ICMP unreachable messages and ICMP fragment needed messages. We conduct tests to evaluate the applicability of these messages in different scenarios to understand the conditions under which our measurement methods can cover targets effectively.

4.1 ICMP Unreachable Message

The applicability of our ICMP unreachable method depends on whether the target host aborts the connection when it receives ICMP unreachable messages. We observe that different operating systems exhibit different action patterns upon receiving ICMP unreachable messages. Since Windows hosts do not accept ICMP messages by default, we mainly tested different versions of Linux. Table 3 shows the results of experiments run in a local web server environment on multiple distributions of different Linux kernels (CentOS, Ubuntu, and Fedora), which are summarized into three categories: older versions of the Linux kernel (from 2.6.32 to 4.0.4), newer versions of the Linux kernel (from 4.5.5 to 6.5.0), and a special 4.2 Linux kernel that doesn't accept any ICMP unreachable messages. Since our method uses a set of ICMP unreachable messages covering all defined codes, it is applicable to all the above systems except 4.2.

However, we notice that the actions taken on ICMP unreachable messages with different error codes do not comply with the recommendations provided in the RFC [8]: The host should not abort the connection upon receiving messages with soft error (i.e. code 0, 1, 5) or undefined code, but should abort the connection when receiving a hard error (i.e. code 2, 3, 4). Such deviations from the RFC recommendations not only cause inconsistencies in how network devices respond to these messages but also potentially degrade the reliability and predictability of network behavior.

4.2 ICMP Fragment Needed Message

To gain insight into the applicability of ICMP fragment needed messages, we test various Linux distribution versions on different kernels in a local network. The results are shown in Table 4 and can be categorized into three groups. (1) Versions 2.6.32 to 3.5.0 of Linux do not check whether the inner packet of an ICMP fragment needed message is legitimate. Instead, they directly update the PMTU values in the routing table. This means that the inner header can be UDP or IP-only packets. (2) Versions 3.0.0 to 3.5.0 of Linux do not update the existing PMTU in the routing table. This means that within the cache validity period, the systems accept only one ICMP fragment needed message pointing to a specific IP. (3) Versions 3.6.0 to 6.5.0 of Linux have the highest PMTU security. They only accept ICMP fragment needed messages if the inner protocol is TCP and the inner 5-tuple matches an existing connection. This means that the first two types of Linux versions could potentially become our measurement targets, as we do not establish connections beforehand. Besides, all versions share the same minimum PMTU values (552 for ICMPv4 and 1280 for ICMPv6) and a cache expiration time of 600 seconds, which ensures that our measurements do not lead to permanent effects.

Notably, the generalizability of our ICMP fragmentation method suggests that a significant number of hosts are still using older versions of operating systems that do not follow the newer RFCs [14][24] that require checking 5-tuples. These hosts are still vulnerable to fragmentation attacks, which can lead to network slowdowns or even complete disruption. Additionally, attackers could exploit these action patterns to identify the target's operating system version and take advantage of vulnerabilities specific to those versions.

5 Measuring Implementation

In this section, we design a comprehensive measurement process for *ICMP_Sonar*. The goal of this section is to realize large-scale ISAV deployment measurements by using the above two methods.

5.1 Finding Measurable Targets

To conduct large-scale measurements, we first need to find sufficient measurable targets, which will take observable action upon receiving the ICMP unreachable or fragment needed messages.

5.1.1 Measurable Targets of ICMP Unreachable Method.

While RFC 1122 [8] specifies that a TCP connection should be aborted upon receiving ICMP unreachable messages, the implementations of some hosts may deviate from the RFC, rendering them unsuitable for subsequent ISAV detection. Therefore, we aim to discover targets that not only support the establishment of TCP connections but also perform retransmissions unless they receive ICMP unreachable messages, which are denoted as measurable targets.

To know if a potential target qualifies as measurable, we perform the following steps: (1) We start by sending a TCP syn to a target. Measurable targets should respond to our probe via TCP syn/ack and perform retransmissions since we do not respond to the TCP syn/ack. Consequently, we should receive n_1 retransmission packets in this step. (2) We send a TCP syn again, and then send a set of ICMP unreachable messages with normal source addresses to the target once we receive the TCP syn/ack. Measurable targets should interrupt the connection process and no longer retransmit. That is, we will receive n_2 retransmission packets in this step. In order to minimize the impact of possible packet loss in the first or second

Linux Kernel	ICMPv4		ICMPv6	
	Abort Connection	Keep Connection	Abort Connection	Keep Connection
2.6.32 - 4.0.4	type=3 / code=0,1,2,3,5-15	type=3 / code=4,u	all	none
4.2	none	all	none	all
4.5.5 - 6.5.0	type=3 / code=0,1	type=3 / code=2-15,u	type=1 / code=1,4-7,u type=4 / code=all	type=1 / code=0,2,3 type=3 / code=all

Table 3: The action of receiving an ICMP unreachable message under different Linux kernel versions. "code=u" indicates a code outside the scope of the definition.

Linux Kernel	ICMPv4		ICMPv6		PMTU Updatable	Cache Expire
	React on Socket	Min PMTU	React on Socket	Min PMTU		
2.6.32 - 2.6.38	no	552	no	1280	yes	600 seconds
3.0.0 - 3.5.0	no		no		no	
3.6.0 - 6.5.0	yes		yes		no	

Table 4: Acceptance conditions for ICMP fragment needed messages, minimum PMTU and cache expiration time for different Linux kernel versions.

step, we consider a target measurable if $n_1 - n_2 \geq 2$, which will be discussed in Section 6.5.

5.1.2 Measurable Targets of ICMP Fragmentation Method.

Some hosts match the inner 5-tuple in the ICMP fragment needed message with an existing connection before the PMTU can be lowered (as discussed in Section 4.2), which cannot be used to detect ISAV in this paper. We need to find target hosts that accept ICMP fragment needed messages without checking.

The ICMP fragmentation method follows similar steps to the ICMP unreachable method for finding measurable targets. (1) First, we send a ping with a payload length of N to potential targets. Measurable targets should reply without fragmentation; otherwise it cannot be distinguished from the fragmented reply of the second step. (2) Second, we send an ICMP fragment needed message before the ping. We construct the payload with a fake IP header where the source IP is assigned as the target host's IP, and the destination address is our IP. This header serves as the payload for the ICMP fragment needed message. If we receive fragmented ping replies, we consider the target as measurable.

5.2 ISAV Detection

After identifying measurable targets, we send spoofed probes to these targets to determine whether they are protected by ISAV. The process of spoofed probing is consistent with the methodology in Section 3. We simply use the neighbor IP address of the target IP address as the spoofed source address.

We take the responses received in the probe as our results. The results of our probing are categorized into the following two categories:

- **Packet intercepted by ISAV**: Normal TCP retransmissions received. / Unfragmented ping reply received.
- **Packet not intercepted by ISAV**: Fewer TCP retransmissions received. / Fragmented ping replies received.

The occurrence of the "Packet intercepted by ISAV" category indicates that there was no change in the target before and after sending the spoofed packet, because the spoofed packet was intercepted by ISAV. Conversely, the "Packet not intercepted by ISAV" indicates that the target received the spoofed packet and took action. In cases where no TCP syn/ack or ping reply is received due to potential packet loss, we will conduct a re-measurement for these targets. Specifically, for the very few targets where measurement results conflict between the two methods, we consider those targets as unmeasurable. We will analyze in detail where and how often such conflicting results or potential measurement errors can occur in Section 6.5.

5.3 Large-Scale Measurement Settings

We are able to conduct large-scale ISAV deployment measurement experiments because the two methods used by our *ICMP_Sonar* have the advantages of simplicity and speed, requiring minimal packets to be sent to the target. Before obtaining a comprehensive view of ISAV deployment across the entire Internet, we need to address a few issues.

Which hosts are we measuring? In IPv4, there are approximately 2^{32} IP addresses, and *ICMP_Sonar* is fully capable of scanning all IPv4 addresses to obtain the most comprehensive measurement results.

However, the vast number of IPv6 addresses makes full scanning impossible. Many previous IPv6 measurements used the existing IPv6 hitlists as measurement targets. We use [35] and [39] as sources for the IPv6 hitlists. One source [35] divides the IPv6 address space into 3 regions based on the number of seed addresses and proposed probing algorithms for each region. The other source [39] constructs the IPv6 hitlist from various sources such as domain lists, FDNS, and others. They both offer regularly updated, publicly accessible services of online active hitlists. During our measurement period, these two hitlists contained 74 348 374 and 9 063 318 active IPv6 addresses, respectively.

How fast should we conduct the measurements? Unlike the approach proposed in [29], our *ICMP_Sonar* does not require sending a large number of packets to reach the ICMP rate limit of the target router. Therefore, we can measure all target hosts

MTU Field	1450	1400	1350	1300	1250
Measurable Targets	262 980	265 435	265 940	266 568	267 766

Table 5: Number of measurable targets in a /8 IPv4 network for different MTU field

with lower probing overhead and a higher probing rate without worrying about adverse effects such as congestion on the target network due to the large number of packets.

However, there may be mutual interference between different steps of experiments on the same target host. In the ICMP target unreachable method, the target host may be unwilling to reset the connection after multiple TCP syn/ack retransmissions, keeping the connection in the SYN-RCVD state for an extended period. In the ICMP fragmentation method, the target host usually caches the set PMTU value in the routing table for 600 seconds. To prevent interference between steps, there is a minimum 10-minute interval between each step of finding measurable targets and detecting ISAV, allowing the connection to be aborted and the PMTU cache in the target host's routing table to expire.

How do we determine the MTU value to be set? Different values of the MTU field (N in Section 3.2) in the ICMP fragmentation method may also affect the measurement results. An MTU value that is too high may exceed the original PMTU on the source-destination forwarding path, resulting in the loss of potential measurable targets. An ICMP fragment needed message with an MTU value that is too low may be rejected by some targets and become ineffective (e.g., the minimum PMTU is restricted to 1280 in IPv6).

To investigate whether the setting of the MTU field will have an impact, we randomly select an IPv4 /8 block to experiment with the different MTU field values. The numbers of measurable targets within the target network are shown in Table 5. The result demonstrates that MTU field values within the range of 1250-1450 have a limited impact (no more than a 2% difference) on the number of measurable targets. Ultimately, we choose 1300 (above the minimum limit of 1280 bytes for IPv6) as our MTU field value for both IPv4 and IPv6.

How do we set the port number? In terms of the ICMP unreachable method, the coverage of the measurement is closely related to the destination port in the TCP syn packet. Only if the corresponding port is open and accepts our connection request can it serve as our potential measurement target. In our study, we select 4 commonly used ports as the destination ports for our measurements, namely ports 22 (SSH), 53 (DNS), 80 (HTTP), and 443 (HTTPS). It is worth noting that our method is not restricted to these ports alone. We can select any open destination port for our measurements.

6 Measurement Results and Analysis

In this section, we first present the number and distribution of measurable targets to demonstrate the broad coverage and low bias of our methods. We then present and analyze the deployment of ISAV across the entire Internet. Finally, we analyze potential errors in our methods, and conduct a survey to investigate the real status of ISAV on the Internet.

6.1 Number and Distribution of Measurable Targets

We deployed *ICMP_Sonar* on three servers within the same network that did not deploy OSAV and conducted large-scale measurements from 2023.12 to 2024.1. We used the two proposed methods to measure the entire IPv4 space and the IPv6 hitlists. Table 6 shows the number of IPv4 and IPv6 targets covered by our measurements. We can see that approximately 86M IPv4 hosts and 0.8M IPv6 hosts were measurable, meaning we could determine whether they were protected by ISAV or not. In contrast, the DNS reflection-based method [28], which is a state-of-the-art approach supporting dual-stack measurements, can only discover 5.3M IPv4 and 15.9K IPv6 measurable targets (i.e. open DNS resolvers). Our proposed method discovers 16.2 (51.6) times more IPv4 (IPv6) measurable targets than the DNS reflection-based method. Most other studies, on the other hand, do not specify a clear number of measurable targets.

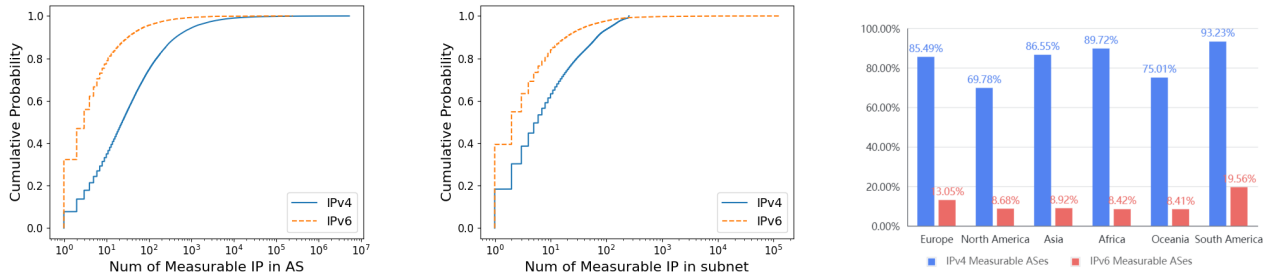
Another thing worth noting is that the overlap of measurable target hosts between the ICMP unreachable method and the ICMP fragmentation method is very small, with only 2 366 829 targets, accounting for just 2.73% of the total measurable targets. This can be explained by the different requirements each method has for the targets, as demonstrated in the results of Section 4. The ICMP unreachable method requires targets that accept connections and support retransmission on certain ports, while the ICMP fragmentation method needs targets that can accept IP-based ICMP fragmentation needed messages, which is supported by older system versions. Because these two methods cover different sets of targets, they complement each other well, resulting in a more comprehensive measurement. This result also indicates that a significant portion of Linux servers running common services (such as HTTP, DNS, etc.) are immune to ICMP fragmentation attacks [23] due to the use of newer system versions.

We also analyze the distribution of measurable targets across different aggregation levels of networks (specific network coverage will be discussed later in Section 6.2). Figure 3a and Figure 3b plot the distribution on the number of measurable targets within each measurable AS and measurable subnet. The results show that 92.29% of ASes and 81.65% /24 subnets in IPv4, as well as 67.67% of ASes and 60.56% /40 subnets in IPv6 have multiple measurable hosts. Our fine-grained measurement methods capable of measuring a sufficient number of target hosts make our results more reliable.

In addition to the significant advantages in quantity, the geographic distribution of the measurable targets is relatively broad and balanced, indicating that our ISAV measurements exhibit low bias. We map each measurable target to a geographic continent based on CAIDA AS organization data [1]. Figure 3c plots the ratio of measurable ASes per continent. There is no excessive disparity in measurable proportions across continents in the figure, which suggests that the distribution of our measurable targets across continents is relatively balanced. The broad and balanced continental coverage of our methodology demonstrates its effectiveness in

Method_Port	Unreach_22	Unreach_53	Unreach_80	Unreach_443	Unreach_ANY	Frag_ANY	Both_ANY
IPv4 Measurable IP	18 209 220	4 416 419	33 675 241	23 320 102	53 361 644	35 687 347	86 682 162
IPv6 Measurable IP	249 363	22 601	312 187	242 420	684 131	143 605	820 768

Table 6: The number of measurable hosts for different methods. 'Port' refers to the destination port of the TCP syn in the ICMP unreachable method.



a CDF of ASes with different numbers of measurable hosts.

b CDF of /24, /40 subnets with different numbers of measurable hosts.

c Ratio of measurable ASes to total ASes for each continent.

Figure 3: Distribution of measurable targets in IPv4 and IPv6.

capturing a comprehensive and representative snapshot of ISAV deployment.

6.2 ISAV Deployment Overview

The measurable targets can be used to verify whether it can receive spoofed packets, i.e., determining whether the network where it is located deploys ISAV. We aggregate the IP-level results and define the following three categories of network-level results:

- **No ISAV**: This category indicates that all measurable hosts within the AS/Subnet are shown to be unprotected by ISAV. The network lacks defense against spoofing attacks.
- **Deployed ISAV**: This category indicates that all measurable hosts within the AS/Subnet are shown to be protected by ISAV. The network is secure in our measurement.
- **Partial ISAV**: This category indicates that measurable hosts within the AS/Subnet show inconsistent measurement results. The network is most likely not deploying ISAV at its boundary but only performing filtering at the ingress of smaller network segments[1]. Thus, these networks can only protect some of the internal hosts from spoofing attacks.

Table 7 presents the aggregated results. The aggregation levels are AS level and Subnet level (/24 subnet in IPv4 and /40 subnet in IPv6). Some networks are categorized as Partial ISAV, when different hosts (measurable in only one of the methods) exhibit different ISAV results. That is why in some cases the number of "both" is lower than the number of separate methods.

Our measurement methods cover a total of 58 891 IPv4 ASes and 8364 IPv6 ASes. Additionally, our methods also cover 3.5M IPv4 /24 subnets and 24K IPv6 /40 subnets. It is important to recognize that the relatively lower network coverage in IPv6 than IPv4 primarily stems from our focused probing on targets within the IPv6 hitlists. Expanding the IPv6 hitlist would likely increase our measurement coverage, yielding a more thorough representation of IPv6 network security.

Compared to previous work, our method achieves quite high coverage in both IPv4 and IPv6, as well as at different granularity levels, including AS level and subnet level. The state-of-the-art ISAV measurement method that supports dual-stack measurements, known as the DNS reflection-based method [28], obtained measurement results with a coverage of 34.8K ASes and 1.2M /24 subnets under IPv4, as well as 4.9K ASes and 10.6K /40 subnets under IPv6. In contrast, our method improves the number of covered ASes and subnets by about 69% and 193% in IPv4, and about 71% and 134% in IPv6, respectively. Although [11] in IPv4 (58K) and [29] in IPv6 (9.7K) have AS-level coverage similar to ours respectively, they can only measure single-stack networks, and the accuracy of their measurements depends on the quietness of the target network. As a comparison, we also discuss the accuracy of our methods in Section 6.5. The coverage can be attributed to the generalizability of our methods for finding sufficient measurable hosts, which enables us to obtain reliable measurement results with a broader coverage and finer granularity.

Our measurement results show that about 60% IPv4 ASes and 46% IPv6 ASes have deployed (or partially deployed) ISAV, which is much higher than previously reported. According to the limited-coverage measurement results of the DNS reflection-based method [28], only 39% of measurable IPv4 ASes and 5% of measurable IPv6 ASes have deployed ISAV. Moreover, the state-of-art IPv4-specific "SMap" method [11] found that less than 20% of the measurable ASes (58K) deploy ISAV, and the state-of-art IPv6-specific method [29] found that about 33% of the measurable ASes (9.7K) deploy ISAV. In addition, the proportion of ASes that partially deploy ISAV in our measurements is relatively higher compared to other studies, with 49% IPv4 ASes and 33% IPv6 ASes, respectively.

[1]A portion of the filtering with very little protection is just user-level NAT [22].

IP Version	Aggregation Level	Method	No ISAV		ISAV		Partial ISAV		Sum
IPv4	AS	unreach	29 414	0.52	6028	0.11	20 891	0.37	56 333
		frag	11 705	0.31	6417	0.17	19 455	0.52	37 577
		both	23 700	0.40	6509	0.11	28 682	0.49	58 891
	Subnet /24	unreach	1 457 263	0.55	939 476	0.36	242 602	0.09	2 639 341
		frag	637 350	0.29	818 354	0.37	728 805	0.33	2 184 509
		both	1 406 663	0.40	1 161 444	0.33	959 344	0.27	3 527 451
	Host	unreach	24 339 229	0.46	29 022 415	0.54	-	-	53 361 644
		frag	13 748 455	0.39	21 938 892	0.61	-	-	35 687 347
		both	36 690 985	0.42	49 991 177	0.58	-	-	86 682 162
IPv6	AS	unreach	4428	0.58	950	0.12	2226	0.29	7604
		frag	1610	0.49	845	0.26	839	0.25	3294
		both	4518	0.54	1115	0.13	2731	0.33	8364
	Subnet /40	unreach	11 506	0.59	3797	0.19	4364	0.22	19 667
		frag	4007	0.43	3584	0.38	1738	0.19	9329
		both	12 775	0.52	6062	0.25	5886	0.24	24 723
	Host	unreach	257 352	0.38	426 779	0.62	-	-	684 131
		frag	82 138	0.57	61 467	0.43	-	-	143 605
		both	334 652	0.41	486 116	0.59	-	-	820 768

Table 7: Aggregated results of large-scale measurements. Each item in the results includes the corresponding number and percentage. The percentage is calculated based on the "Sum" column, which refers to the sum of the three categories of network-level results in each row. Besides, the results of the ICMP target unreachable method and the ICMP fragmentation method are displayed on separate rows, while "both" shows the union of the results from both methods.

Measurement Targets	Time	No ISAV		ISAV		Partial ISAV		Sum
DNS-reflection	2020	21 314	0.61	2092	0.06	11 441	0.33	34 847
Our DNS	2024	20 095	0.54	4729	0.13	12 048	0.33	36 872
Our Total	2024	23 700	0.40	6509	0.11	28 682	0.49	58 891

Table 8: Comparison of AS-level results with DNS-reflection method in IPv4. "Our DNS" refers to the measurement results of our targets that are also measurable to the DNS-reflection method, while "Our Total" represents the results for all our targets.

We use the numerical comparison with the the DNS reflection-based method [28] as an example to explain the possible reasons for such a significant discrepancy in measurement results. We filter out the 4.4M IPv4 measurable targets in our dataset that have port 53 open, which should represent the upper limit of the number of targets in our dataset that meet their method's requirements (i.e. open DNS resolvers). The remaining 82M measurable targets are exclusive to our approaches. The aggregated AS-level results for these targets are listed in Table 8. According to this table, the measurement results from the 4.4M targets reveal that 54% of ASes have not deployed ISAV, while the remaining 13%/33% of ASes have fully/partially deployed ISAV. Compared to the overall measurement results reported in [28], there are only a small part of ASes that did not deploy ISAV have deployed ISAV over the years, while the proportion of ASes with partial ISAV deployment has remained unchanged. In addition, we also analyze the number of measurable hosts in each AS for those limited targets, and only 47.86% of ASes have more than one measurable host (significantly lower than 92.29% of ASes for all targets in Figure 3a). The limited or biased measurable targets may result in misclassifying partially deployed ISAV as deployed or not deployed. In contrast to the measurements from the limited targets, our 82M exclusive measurable targets help identify a completely different result, i.e. the number of ASes with ISAV partially deployed is much higher compared to their measurements. Therefore, the widely distributed measurable targets (as shown in Figure 3) of our methods can help identify ASes with partial ISAV more accurately, i.e., capturing subnet-level deployment strategies.

6.3 Deployment Status of ISAV

In the following, we analyze the measured results of ISAV deployments to detail how different ASes implement ISAV and the factors that influence deployment practices.

6.3.1 Network administrators tend to deploy ISAV at the boundaries of smaller-scale networks rather than at the boundaries of ASes.

The large proportion of ASes with partial ISAV indicates that the ISAV deployment status varies among different subnets within many ASes. This may be attributed to constraints in network topology, routing policies, or administrative factors. Moreover, we also find that a much larger proportion (27.2% for IPv4 and 23.8% for IPv6) of measurable subnets exhibit partial ISAV deployment compared to previous results (14.4% for IPv4 and 1.2% for IPv6) based on the DNS reflection-based method [28]. This indicates that ISAV might be deployed in networks smaller than /24 IPv4 subnets and /40 IPv6 subnets.

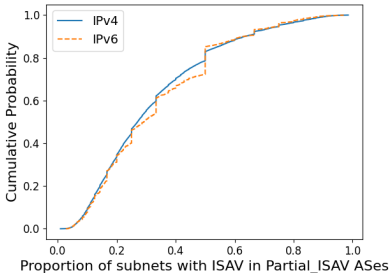


Figure 4: CDF graph of proportion of subnets that deployed ISAV in PARTIAL_ISAV ASes.

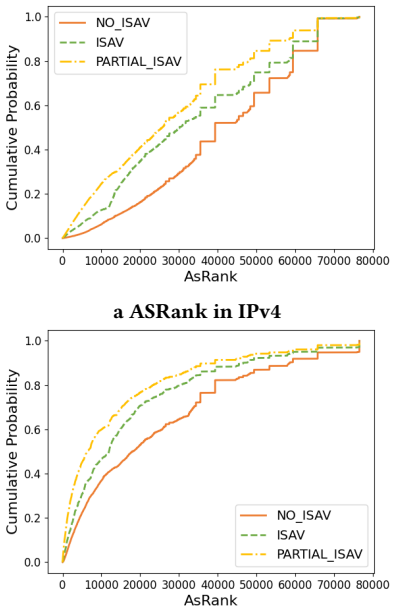
In addition, we plot the CDF on the proportion of subnets that deployed ISAV per PARTIAL_ISAV AS (see Figure 4), which serves as an indicator of the security ratio of these PARTIAL_ISAV ASes. We observe that over 80% of the PARTIAL_ISAV ASes have less than half of their subnets deploying ISAV. This result suggests that while these ASes have initiated measures to implement ISAV, the coverage within their networks is significantly incomplete, which may not effectively mitigate the risks associated with IP spoofing. The low subnet deployment rates within these PARTIAL_ISAV ASes indicate that fine-grained measurement and analysis better reflect the actual situation of ISAV deployment.

6.3.2 Networks of different levels, locations, and categories exhibit significant differences in ISAV deployment.

AS tier and AS rank: AS tier [16] represents its position in the Internet hierarchy. Higher-tier ASes (e.g. clique) are closer to the core of the Internet, while lower-tier ASes (e.g. stub tier) are closer to the edge. Table 9 illustrates the disparities in ISAV deployment across different AS tiers. Higher-tier ASes in both IPv4 and IPv6 networks exhibit a higher deployment rate of ISAV, including both deployed and partially deployed implementations. In contrast, stub tier ASes have a notably lower rate of ISAV deployment compared to other tiers. Taking the IPv4 network as an example, the deployment rate in clique ASes is 89%, which is 62% higher than that of stub-tier ASes (55%). Moreover, higher-tier ASes are more likely to implement subnet-level deployment strategies, i.e., partial ISAV.

The trend is also reflected in ASRank [2], which is established by CAIDA based on the customer cone size of ASes. We obtained the latest ASRank dataset from CAIDA in 2024.02 and compared it with our ISAV measurement results. Figure 5 plots the distribution of ISAV deployment across different AS rankings. We can see that under both IPv4 and IPv6 networks, ASes with "ISAV" and "PARTIAL_ISAV" are predominantly concentrated at higher-ranking levels. This trend indicates that ASes with larger scale and greater influence are more likely to implement ISAV-related security measures. In contrast, lower-rank ASes might relatively lack awareness of security defenses against spoofing attacks.

AS type: ASes categorized as different types may also adopt distinct ISAV deployment practices. We obtain AS type data from the Stanford ASdb Dataset [40] and list the ISAV deployment status for typical AS types in Figure 6. There are clear differences in ISAV deployment across different AS types. "Computer and Information Technology" ASes show significantly better deployment rates in



a ASRank in IPv4

b ASRank in IPv6

Figure 5: CDF graph of ISAV deployment across different ASRanks.

IPv4 compared to other categories. This result is expected, as IT companies or ISPs should have a better understanding of spoofing attacks and SAV. In contrast, "Education and Research" ASes perform relatively better in IPv6 deployment. This could be due to the fact that a significant part of the current IPv6 development is being driven by the academic community.

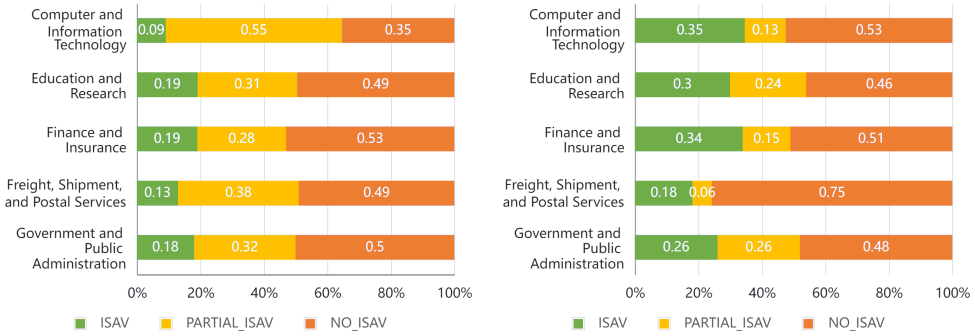
Geographic location: To further explore the differences in ISAV deployment across different geographic locations, we map the measurable targets to countries and plot a heatmap that illustrates the level of spoof vulnerability across countries (see Figure 7). Differences in the rate of (full or partial) ISAV deployment across countries are not significant in IPv4. However, in IPv6, the deployment rate varies widely from country to country. Some countries in Africa and South America exhibit poorer ISAV deployment rates, making their networks more susceptible to attacks. This disparity highlights the need for a more globally coordinated effort to enhance ISAV implementation and bolster overall cybersecurity defenses against spoofing attacks.

6.3.3 Networks differ significantly in their ISAV deployments under IPv4 and IPv6.

Currently, the status of ISAV deployment is better in IPv4 compared to IPv6. The data in Table 7 show that 40% of measurable IPv4 ASes have not deployed ISAV at all, whereas in IPv6, this figure rises to 54%. Figure 7 provides a more visual representation of differences in the geographic distribution of security against spoofing attacks for IPv4 and IPv6. We also compare the discrepancies in ISAV deployment within the same ASes: Only 49% of networks that have deployed (or partially deployed) ISAV in IPv4 have also deployed ISAV in IPv6; in contrast, 92% of networks that have deployed (or

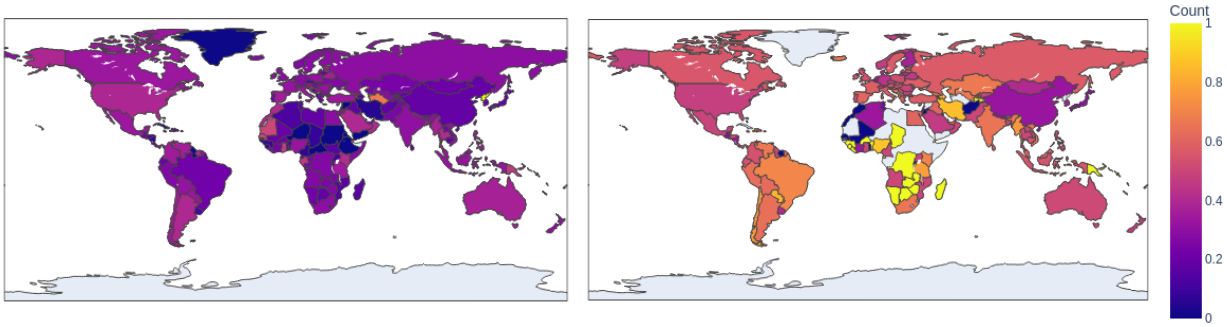
IP Version	Tier	No ISAV		ISAV		Partial ISAV		Sum
IPv4	clique	2	0.11	1	0.06	15	0.83	18
	high tier	179	0.13	199	0.15	951	0.72	1329
	low tier	1865	0.19	755	0.08	7130	0.73	9750
	stub tier	21 327	0.45	5459	0.12	20 281	0.43	47 067
IPv6	clique	1	0.06	1	0.06	16	0.88	18
	high tier	240	0.27	150	0.17	488	0.56	878
	low tier	1625	0.50	412	0.13	1216	0.37	3253
	stub tier	2409	0.62	514	0.13	951	0.25	3874

Table 9: AS Tier distribution of ISAV measurable ASes. "Sum" is the sum of the first three items, representing the number of measurable ASes.



a ISAV deployment in IPv4 **b ISAV deployment in IPv6**

Figure 6: ISAV deployment status for typical AS types.



a IPv4 world heatmap **b IPv6 world heatmap**

Figure 7: World heatmap of spoof vulnerability. Brighter colors indicate higher vulnerability.

partially deployed) ISAV in IPv6 have also deployed ISAV in IPv4. This may indicate that network administrators, when deploying SAV, focus more on the security of IPv4 and selectively neglect IPv6.

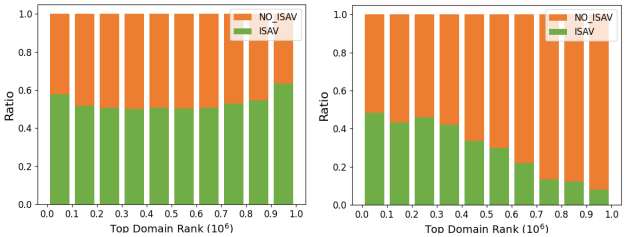
6.4 Case Study: ISAV Status of Top Domain

To illustrate the impact of missing ISAV deployments on critical network infrastructure, we analyze the extent to which top domains are vulnerable to spoofing. Top domains are particularly important because they often host critical services such as commerce, government, and education, and tend to suffer from spoofing attacks. Deploying ISAV for these domains is critical to maintaining the security and reliability of the Internet.

We obtained the Tranco top 1 million domain dataset [30] in 2024.03 as our measurement targets. We acquire IP addresses for these domains through DNS queries and then conduct measurements on these IP addresses. The measurement results for these top domains are shown in Table 10. We obtain measurement results for 205 124 IPv4 domains and 9428 IPv6 domains. Even for such critical infrastructure as the top 1M domains, only 54% of measurable IPv4 domains and 16% of measurable IPv6 domains are protected.

IP Version	NO_ISAV		ISAV	
IPv4	95 064	0.46	110 060	0.54
IPv6	7887	0.84	1541	0.16

Table 10: ISAV deployment of Tranco top 1M domains



a Top domain with IPv4 IP **b Top domain with IPv6 IP**

Figure 8: ISAV deployment rates of top domains under different domain ranks

To investigate whether domains with higher popularity are more likely to be protected by ISAV, we divide the top 1M domains equally into 10 sets according to their popularity rankings and plot the ISAV deployment ratio of each set in Figure 8. We observe that the top 400 000 domains exhibit similar ISAV deployment rates in both IPv4 and IPv6 networks. As for the remaining domains, lower-ranked domains show progressively lower ISAV deployment rates for IPv6 networks, while in IPv4 networks, the deployment rates remain relatively consistent across all popularity ranks. This suggests that networks hosting higher-ranked domains place a similar emphasis on the security for both IPv4 and IPv6, whereas networks hosting lower-ranked domains do not maintain this consistency, i.e. placing less emphasis on IPv6 domain security than on IPv4.

However, even for the IPv4 top 1M domains, the ISAV deployment rate is still far from satisfactory. To further explore the willingness of ASes to protect top domains from spoofing attacks, we compare the security status of top domains with the security status of the ASes in which these domains reside. We discover that among the 7413 IPv4 ASes (339 in IPv6) with partial ISAV that contain the measurable top domains, 83% IPv4 ASes (80% in IPv6) include at least one top domain not protected by ISAV, and 54% IPv4 ASes (56% in IPv6) fail to protect any top domain at all. This indicates that the partial ISAV deployment strategy is relatively unsafe, as most ASes with partial ISAV cannot successfully secure their critical network infrastructure. More optimal subnet-level ISAV deployment strategies still need further exploration.

6.5 Potential Measurement Error Analysis

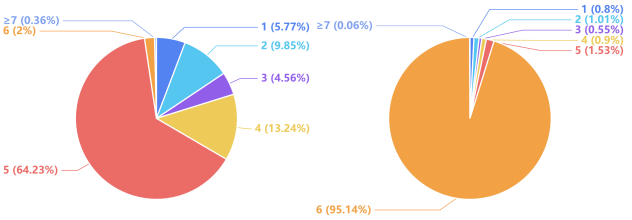
Our two methods, the ICMP unreachable method and the ICMP fragmentation method, tend to produce fewer errors compared to some other techniques. This is because some methods, such as those relying on AS relationship inference (e.g., methods based on traceroute loops [21]), depend heavily on the accuracy of existing AS relationship datasets. If these datasets are outdated or incorrect, the reliability of the measurements can be compromised. On the other hand, methods like those based on IP-ID [11] or ICMPv6 Rate Limiting [29] require the target network to be relatively quiet. In active network environments, these methods may struggle to detect subtle anomalies because the small number of spoofed packets they send may fail to cause a detectable change in IP-ID sequences or ICMPv6 response rates. As a result, these methods can lead to incorrect assessments of SAV deployment. In contrast, our ICMP-based methods are less dependent on external datasets and network quietness.

Besides, our method exhibits a certain resilience to packet loss caused by factors such as host configuration or firewall settings. For the ICMP fragmentation method, in most cases, packet loss results merely in the absence of a ping response, prompting us to re-measure the target. As for the ICMP unreachable method, we analyze the number of TCP retransmissions before and after ICMP unreachable messages are received in large-scale measurements, i.e., the value of n_1 and n_2 defined in Section 5.1.1. Errors in both n_1 and n_2 due to packet loss can lead to incorrect target identification. Therefore, it is critical to set a suitable threshold for the difference between n_1 and n_2 to determine the measurable hosts. The larger the threshold, the less likely the method is to be affected by a false low of n_2 because of packet loss. However, setting the threshold too large may miss some measurable targets. To determine an appropriate threshold, we explore the distribution of $n_1 - n_2$ in large-scale measurements (Figure 9). We observe that, in IPv4, 94.23% of the measurable hosts have a count of TCP retransmission difference of more than one. Meanwhile, the proportion rises to 99.2% in IPv6, with the vast majority concentrating on 6 retransmissions. Therefore, we set the threshold of $n_1 - n_2$ to 2 to minimize the effect of false low of n_2, while maintaining a large amount of measurable targets. This resilience is crucial, as it ensures that our measurement accuracy is maintained despite the common network issues that can affect data transmission.

Additionally, for the ICMP unreachable method, the delayed arrival of ICMP unreachable messages can produce errors that need to be taken into account. The issue of the ICMP unreachable method stems from the requirement that we must receive a response packet from the target and then take it as the inner packet of our ICMP unreachable message. If the target's retransmission interval exceeds the round-trip time (RTT) of packet transmission, the target may only receive the ICMP unreachable message after initiating a new retransmission. This timing discrepancy will lead to the mistaken conclusion that the target did not receive the ICMP unreachable message, potentially misclassifying the target's behavior. To verify the likelihood of such delayed errors, we analyze the time intervals of TCP retransmissions for the targets. We find that 96.79% of the IPv4 target hosts and 99.45% of the IPv6 target hosts wait for 1000ms or more before the first retransmission, while 97.83% RTTs for TCP in our measurement fall below 1000ms. The analysis results indicate that our ICMP unreachable method has a high probability of avoiding errors caused by delay and packet loss.

6.6 Survey on ISAV deployment

To gain insights into the practical considerations of network administrators when deploying SAV, we sent a survey to AS administrators via email. The specific content of the survey can be found in the appendix C. In total, we received 32 valid responses from different



a Distribution in IPv4. **b Distribution in IPv6.**

Figure 9: Distribution of the difference in the number of TCP retransmissions before and after ICMP unreachable messages are received by targets in the large-scale measurements.

ASes, of which 20 ASes have partially or fully deployed SAV, and the remaining 12 ASes have not deployed SAV.

All 20 ASes claim to have deployed SAV in IPv4, while only 17 of them also have deployed it in IPv6, which is consistent with our analysis in Section 6.3.3. With respect to the location of deployment, 13 ASes have deployed SAV at subnet boundaries for reasons including: 1. Ease of management, or the administrator's management scope is limited to subnets. 2. Selective deployment of SAV in certain subnets of the AS due to reasons like multi-homing in some subnets. 3. To regulate traffic between subnets. Regarding the specific implementation of SAV, 9 ASes have solely used ACL (Access Control List) to filter spoofed traffic, without employing uRPF (Unicast Reverse Path Forwarding), which is considered more flexible and effective in large ISPs [36][6].

The survey results indicate that the current deployment of SAV varies and is complex both among different ASes and within individual ASes. Only with sufficient coverage and fine-grained measurements as well as analysis that penetrate deep into ASes can we gain a comprehensive overview of the actual deployment of SAV across the entire Internet.

7 Conclusion

Our study on the measurement of ISAV deployment using meticulously crafted spoofed ICMP messages has unveiled significant insights and raised urgent discussions about ISAV deployment practices. Our ISAV deployment measurement approach surpasses previous research in dual-stack capability, high coverage and accuracy, and fine granularity.

Our measurements reveal that the proportion of networks with partial ISAV deployment is significantly higher than reported in previous studies. ISAV deployment varies significantly across different network boundaries, levels, locations, and between IPv4 and IPv6, with a tendency to be implemented more at the edges of smaller networks rather than AS boundaries. The experiments also show that critical infrastructure, like top domains, is not prioritized for protection over other networks, highlighting that partial ISAV deployment is insufficient for ensuring comprehensive security. These findings serve as a caution to SAV researchers and network administrators, suggesting that the deployment of SAV is still far from sufficient.

While our study has provided valuable insights, it also highlights areas for future improvement. First, the hitlists limit the breadth of our measurement coverage in IPv6. Second, our methods focus only on whether the target network intercepts the spoofed packets we send, without delving into the specific mechanisms of SAV deployed by the targets (such as ACL or uRPF). These are issues we hope to address in our future work.

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A Ethics

Our research strictly adheres to the ethics of network measurement, aiming to minimize the impact on target systems while securely assessing their ISAV deployment. We will describe our ethical efforts in the following areas:

Measurement Targets: We conduct equivalent measurements across the complete IPv4 address space and all targets within two IPv6 hitlists. Our approach is non-discriminatory with respect to specific countries, industries, or address spaces.

Measurement Rate: We meticulously control the quantity and rate of packets sent (no more than 10 packets per target host every 10 minutes) to prevent overloading the target networks. This precaution helps to minimize any potential disruption or degradation of service on the target systems while conducting our measurements.

Potential risks of ICMP error messages: In our measurement, ICMP unreachable messages are only used to interrupt the upcoming connection between the target host and our machine, and do not affect any other normal communication of the target host. ICMP fragment needed messages can reduce the PMTU in the target host's routing table. However, our method is safe because it controls the target to reduce the PMTU only related to our IP and does not affect other communications. Additionally, we set a reasonable PMTU value (1300 bytes, as detailed in Section 5.3), and this PMTU will be restored after the cache expires (usually 600s, as detailed in Section 4.2) without causing a lasting impact.

Due to the sensitivity of ISAV deployment results, which can reveal vulnerabilities to spoofing attacks in specific networks, we take a cautious approach in presenting and analyzing the data. We only display and analyze aggregated results to avoid exposing any particular network's weaknesses publicly. Detailed ISAV deployment results are disclosed exclusively to network administrators of the respective networks.

B ICMP Unreachable Error Type and Code

See Table 11.

Version	Type	Code	Description
ICMPv4	3	0	Network unreachable
		1	Host unreachable
		2	Protocol unreachable
		3	Port unreachable
		4	Fragmentation unreachable
		5	Source route failed
		6	Destination network unknown
		7	Destination host unknown
		8	Source host isolated
		9	Network administratively prohibited
		10	Host administratively prohibited
		11	Network unreachable for TOS
		12	Host unreachable for TOS
		13	Communication administratively filtered
		14	Host precedence violation
		15	Precedence cutoff in effect
ICMPv6	1	0	No route to destination
		1	Administratively prohibited
		2	Beyond scope of source address
		3	Address unreachable
		4	Port unreachable
		5	Source address failed ingress/egress policy
		6	Reject route to destination
		7	Error in Source Routing Header
	3	0	Hop limit exceeded in transit
		1	Fragment reassembly time exceeded
	4	0	Erroneous header field encountered
		1	Unrecognized Next Header type encountered
		2	Unrecognized IPv6 option encountered
		3	First Fragment has incomplete Header Chain

Table 11: ICMP Unreachable message types, codes, and descriptions

C Questionnaire

Q1: Your AS Number.

Q2: Name of Organization.

Q3: Did you know about Source Address Validation (SAV) before?

(1) Yes
(2) No

Q4: Have you implemented Filter or Source Address Validation (SAV) in your network?

(1) Yes (To Q7)
(2) No
(3) Unknown

Q5: Are there any limitations or concerns that have impacted your SAV deployment?

Q6: Are you planning to implement SAV in the future? (To Q13)

(1) Yes
(2) No
(3) Unsure

Q7: Have you implemented SAV in both IPv4 and IPv6?

(1) Both in IPv4 and IPv6
(2) Only in IPv4
(3) Only in IPv6

Q8: Do you filter outbound or inbound traffic?

(1) Only outbound filtering
(2) Only inbound filtering
(3) Both
(4) Unsure / Auto Configuration

Q9: Where have you deployed SAV?

(1) At the AS boundary
(2) At subnet boundaries within the AS
(3) Both AS and subnet boundaries
(4) Other (please specify)

Q10: What are the reasons you chose to deploy here?

Q11: What types of SAV filtering techniques are you using?

(1) Access Control List (ACL)
(2) Strict Unicast Reverse Path Forwarding (Strict uRPF)
(3) Loose Unicast Reverse Path Forwarding (Loose uRPF)
(4) Feasible Path Unicast Reverse Path Forwarding (Feasible uRPF)
(5) Enhanced Feasible Path Unicast Reverse Path Forwarding (EFP-uRPF)
(6) Other (please specify)

Q12: What challenges have you encountered in implementing SAV?

Q13: How effective do you believe SAV is in mitigating IP spoofing and DoS attacks in networks? (1-10)

Q14: We are conducting a large-scale study on SAV deployment probing. Would you be interested in receiving the results for your network in the future?

(1) Yes, my email is _____
(2) No, thanks